Methods	Core Algorithm/Model	How to estimate kinetic parameters		Whether to calculate and output du/dt	Performance at Cell level					Performance at Gene level				
					Local		Global			Gene-level kinetics			Kinetic parameters	
		α	β&γ		consistency score	ICCoh	CBDir	TransCosine	BATC	Real_CBDir	Sim_CBDir	Sim_CosSim	α_Spearman	γ_Spearman
scVelo_deterministic (Velocyto model)	Steady-state Linear Regression	——	Constant(β) Gene-specific (γ)	——						——	——	——	——	——
scVelo_stochastic	Stochastic Model (Second-order Moments Approximation)	——	Constant(β) Gene-specific (γ)	——						——	——	——	——	——
scVelo_dynamic	Dynamical Model (EM Algorithm with Analytical Solution)	Discrete States	Gene-specific	Explicit Output									——	——
Multivelo_dynamic (RNA_only)	Coupled ODEs (Chromatin-RNA interactions)(c=1)	Discrete States	Gene-specific	Explicit Output									——	——
UniTVelo	Unified Time Model	Gene-specific (RBF model)	Gene-specific	Yes, not output									——	——
UniTVelo(mode2)	Independent Time Model	Gene-specific (RBF model)	Gene-specific	Yes, not output						——	——	——	——	——
Dynamo (RNA_seq)	RNA velocity estimation + SparseVFC	Constant	β fixed γ gene-specific	——						——	——	——		——
TIvelo	Trajectory Inference Guidance + Orientation Score	——	——	Explicit Output									——	——
cell2fate	Bayesian Linearized ODE (implemented in Pyro)	Linear combination of modules	Gene-specific	Yes, not output						——	——	——	——	——
TFvelo	Gene Regulatory Network (GRN) + Generalized EM Algorithm	Linear combination of TFs	——	——						——	——	——	——	——
veloAE	Autoencoder Representation Learning	——	β fixed latent-space γ	——						——	——	——	——	——
veloVI	Deep Generative Model (VAE)	Discrete States	Gene-specific	Explicit Output									——	——
DeepKINET	Deep Generative Kinetic Model	——	Cell-specific (VAE Output)	——						——	——	——	——	
TSvelo	Neural Ordinary Differential Equation (Neural ODE)	Continuous function α(t)	Gene-specific	Explicit Output						——	——	——		——
cellDancer	Deep Neural Network (DNN)	Cell-specific	Cell-specific (DNN Output)	Explicit Output										
Deepvelo(GCN)	Graph Convolutional Network (GCN)	Cell-specific	Cell-specific (GCN Output)	Explicit Output										
SDEvelo	Stochastic Differential Equation (SDE)	Sigmoid function	Gene-specific	Yes, not output						——	——	——	——	——
Deepvelo_scVelo	Neural ODE + VAE (Requires existing velocity estimates)	——	——	from scVelo									——	——
Graphvelo_scVelo	Tangent Space Projection (TSP) + kNN-based Velocity Correction	——	——	from scVelo									——	——
Graphvelo_Dynamo	Tangent Space Projection (TSP) + kNN-based Velocity Correction	——	——	——						——	——	——	——	——